

Developer Build Brief

Company Brain — Decision Harness

Hackathon: Multi-Agent Orchestration Build Day, May 31 2026

What we're building

A multi-agent decision orchestrator. Five specialist agents evaluate a question independently. An orchestrator reads all five, detects conflict, applies weighted scoring, and produces a structured verdict. Full W&B Weave instrumentation throughout. Human-in-the-loop override panel lets judges adjust weights and inject context, then re-run the orchestrator live.

Demo question: "Should this project win the Most Sophisticated Harness prize?"

Stack

- Python 3.11+
 - `openai` (or `anthropic`) — LLM calls
 - `weave` — instrumentation and tracing
 - React frontend (already built — see `demo_v2.jsx`)
 - FastAPI — one endpoint connecting frontend to Python backend
 - No database. No auth. No persistence needed.
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Build tasks — in priority order

BLOCK 1 — Core harness (you must finish this)

Estimated: 2.5 hours

- ☐ `pip install weave openai fastapi uvicorn` — 5 min
- ☐ `weave.init("company-brain-harness")` at top of `main.py` — 2 min
- ☐ Write 5 agent functions, each decorated with `@weave.op()` — 60 min
 - Each function takes: question (str), returns: dict (JSON)
 - System prompts are provided — just paste them in
 - Use gpt-4o or claude-sonnet

- ☐ Write orchestrator function, also `@weave.op()` — 30 min
 - Takes: dict of 5 agent outputs + weights dict + override string
 - Returns: structured verdict dict
 - Orchestrator prompt provided — paste it in
 - ☐ Run all 5 agents, pass outputs to orchestrator, print result — 20 min
 - Confirm it runs end to end before anything else
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BLOCK 2 — Weave instrumentation (required to win)

Estimated: 1 hour

- ☐ Add scorer for numeric confidence — 15 min

```
python

@weave.op()
def score_confidence(output: dict) -> float:
    return output.get("score", 0) / 10
```

- ☐ Add scorer for verdict strength — 15 min

```
python

@weave.op()
def score_verdict_strength(output: dict) -> float:
    return {"STRONG": 1.0, "CONDITIONAL": 0.5, "WEAK": 0.0}.get(output.get("verdict"))
```

- ☐ Run weave.Evaluation with both scorers across all 5 agents — 20 min
 - ☐ Get shareable Weave URL — paste into submission — 10 min
 - Run once, go to wandb.ai, find the project, copy the URL
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BLOCK 3 — FastAPI backend (connects frontend to Python)

Estimated: 45 min

- ☐ One POST endpoint `/run` — 30 min

```
python
```

```
@app.post("/run")
async def run(payload: dict):
    question = payload["question"]
    weights = payload.get("weights", default_weights)
    override = payload.get("override", "")
    # run agents, run orchestrator, return results
    return {"agents": agent_results, "orchestrator": orch_result}
```

- ☐ CORS enabled (frontend needs it) — 5 min
 - ☐ Test with curl before connecting frontend — 10 min
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BLOCK 4 — Frontend connection (only after Block 3 works)

Estimated: 30 min

- ☐ React frontend is already built (demo_v2.jsx)
 - ☐ Update fetch URL from Anthropic API to `http://localhost:8000/run`
 - ☐ Pass weights and override text from UI state to POST body
 - ☐ Confirm full flow works: button click → agents run → verdict appears
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BLOCK 5 — Polish (only if time allows)

Estimated: 30 min

- ☐ Run same question twice with different weights — save both in Weave
 - ☐ Show side-by-side comparison in Weave dashboard
 - ☐ Add agent org names to Weave trace metadata
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What's already done — do not rebuild

- All 5 agent system prompts — in hackathon_brief.md
 - Orchestrator prompt — in hackathon_brief.md
 - React frontend UI — demo_v2.jsx (fully functional, calls Claude API directly)
 - Weighted scoring logic — defined in brief
 - 3-minute demo script — in brief
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Critical path

If you run out of time, stop in this order:

1. Block 1 complete + Weave traces visible → minimum viable submission
2. Block 2 complete → scorers in dashboard → strong submission
3. Block 3+4 complete → live demo from UI → winning submission
4. Block 5 → comparison view → exceptional submission

Do not start Block 3 until Block 2 is working. Do not start Block 5 under any circumstances until Block 4 is done.

Submission checklist (due 8pm)

- ☐ Public GitHub repo with all code
 - ☐ Weave project URL (shareable trace link)
 - ☐ Screen recording under 2 minutes
 - ☐ Project description (copy from hackathon_brief.md)
 - ☐ All team member names, emails, socials
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API keys

Request free credits at: <https://forms.gle/rj1tGQK6cBeYmvq38> W&B team is on-site — find them if any Weave questions. WiFi: TheEngine-Guest / toughtech